

Unmanned Aerial Vehicle (UAV)-Assisted Path Planning for Unmanned Ground Vehicles (UGVs) via Disciplined Convex-Concave Programming

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Abstract—The last decade has witnessed growing research interests in the path-planning problem for unmanned ground vehicles (UGVs) equipped with RGB-D cameras or LiDARs. However, apart from their extremely high cost, these sensors may be obscured by obstacles, which makes them impractical for many complex scenarios. Besides, most of the existing works for path planning have not considered energy efficiency for UGVs that are usually constrained by limited on-board batteries. Distinct from existing methods, this work presents a vision-based unmanned aerial vehicle (UAV)-assisted cooperative system for multiple UGVs. The proposed system harnesses the broad vision of UAV and operates in both general outdoor and global positioning system (GPS)-denied indoor environments. In sharp contrast to the conventional heuristic algorithms such as the rapidly exploring random tree (RRT) algorithm and the Dijkstra methods, the proposed energy-efficient path planning for UGVs is formulated as a non-convex optimization problem by considering the collision-aware obstacle-avoidance. More specifically, the proposed path-planning scheme contains two stages, namely the semantic segmentation that localizes UGVs and obstacles in the environment by exploiting the wide-angle camera mounted on the UAV, followed by the trajectory generation in which a disciplined convex and concave programming (DCCP) algorithm is devised for the non-convex energy-minimization problem. Extensive experiments validate the effectiveness of the UAV-UGV cooperative system and the proposed DCCP-based path-planning scheme.

Index Terms—Disciplined convex-concave programming, path planning, semantic segmentation, UAV-UGV cooperative system.

I. INTRODUCTION

PATH planning, as a critical component for intelligent mobile robots and automatic pilots, is expected to identify

the possible collisions before generating energy-efficient trajectories to avoid obstacles for unmanned ground vehicles (UGVs). For instance, UGVs serving as a community policing should have the capability of recognizing stochastic obstacles scattered on the road. Besides, meticulous path planning for collision avoidance among multiple UGVs is also highly demanded. For a UGV to detect obstacles and the other UGVs, range- and appearance-based methods are the two major solutions by applying LiDAR [1] and RGB-D camera [2], respectively. Unfortunately, the performance of these two solutions is not always satisfactory. First, both RGB-D and LiDAR sensors are expensive, which incurs high costs if they are equipped on each UGV individually. Second but importantly, RGB-D and LiDAR sensors suffer from a limited view angle when applied to scenarios of large-scale obstacles [3], [4]. To cope with these challenges aforementioned, unmanned aerial vehicles (UAVs) have been recently proposed as an efficient solution to this problem [5]. This UAV-assisted approach outperforms existing methods in the following aspects:

- *Low Cost*: A typical commercial UAV of less than US\$3000 is capable of serving multiple UGVs simultaneously;¹
- *Long-range Detection*: A UAV with a wide-angle camera can detect objects from a far distance. For instance, the DJI MAVIC2 can achieve the furthest detection range of 180 meters using its built-in camera;²
- *Real-time Communications*: The video stream and photos taken by a UAV can be transmitted back in real-time, which supports latency-sensitive applications such as automatic piloting;
- *Line-of-sight (LoS) Transmissions*: The wireless channel between the UAV and UGVs under coverage is LoS at most time, which enables efficient wireless data communications.

Motivated by the advantages above, this work proposes an efficient UAV-UGV cooperative system for path planning available in both indoor and outdoor environments. With the assistance of one UAV, a path-planning algorithm is developed for multiple

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¹[Online]: Available. https://store.dji.com/shop/mavic-series?from=menu_icon

²[Online]: Available. <https://www.dji.com/products/compare-consumer-drones>

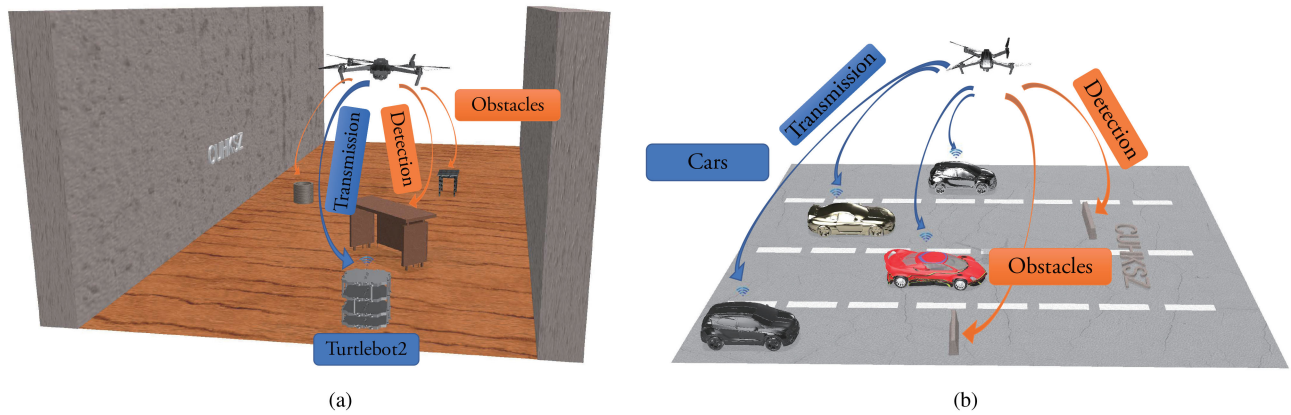


Fig. 1. Illustration of the proposed UAV-UGV cooperative system for indoor and outdoor scenarios. (a) Obstacle-avoidance system for a single UGV. (b) Collision-aware obstacle-avoidance system for multiple UGVs.

UGVs in this work. Unlike [6], this work takes into account the detection of UGVs as well as collision avoidance among UGVs. More specifically, the proposed algorithm is divided into two stages, namely the semantic segmentation and the trajectory generation. In the first stage, obstacles and UGVs can be identified and localized using semantic segmentation techniques by exploiting the wide-view camera mounted on the UAV. For the second stage of trajectory generation, the path-planning problem is mathematically formulated as a non-convex optimization problem before the disciplined convex and concave programming (DCCP) technique is applied to solve the problem [7]. In sharp contrast to the conventional heuristic algorithms such as machine learning (ML)-based methods, the proposed path planning algorithm aims to develop a non-convex optimization problem in which the potential collisions among obstacles and UGVs are explicitly modeled and resolved. In particular, the proposed algorithm does not require a training procedure. As a result, this low-cost and robust algorithm can be practically implemented in real-life applications such as post-earthquake rescues in indoor environment [8].

Fig. 1 illustrates the proposed UAV-UGV cooperative system in which multiple UGVs (cars or robots) are served by one UAV in indoor and outdoor scenarios. The UAV equipped with a camera takes photos and videos to capture the ground UGVs and obstacles. For the outdoor scenarios, the global position can be directly acquired by the GPS equipped on the UAV. In contrast, additional sensors such as TFmini Plus can be equipped on the UAV to obtain the height information in the GPS-denied indoor scenarios. Furthermore, collision and obstacle avoidance are considered in generating the UGV trajectories as shown in Fig. 2. More specifically, given an input image captured by the UAV, the objects are first extracted using semantic segmentation. After that, the detected objects are represented by circular approximation, which enables the path-planning optimization problem to be mathematically formulated as a convex-concave optimization problem. Finally, the energy-efficient trajectory is generated using a DCCP-based algorithm by solving the non-convex optimization problem.

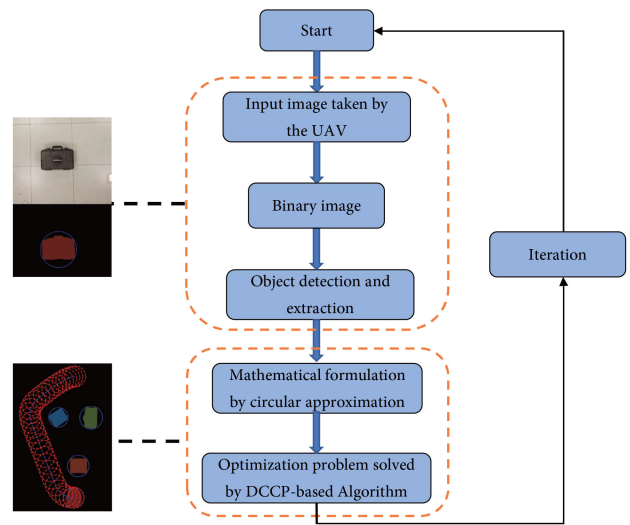


Fig. 2. Flowchart of the proposed algorithm. In the first step, the objects are detected and segmented. Next, the problem is mathematically formulated and solved using DCCP-based algorithm.

In summary, the main contributions of this paper are summarized as follows:

- A UAV-UGV cooperative system consisted of one UAV and multiple UGVs is constructed, operating under the indoor and outdoor environments. In contrast to RGB-D and LiDAR-based systems, fewer sensors are required for UGVs in the proposed system. One UAV is sufficient to serve multiple UGVs at the same time.
- A novel shape representation scheme for detected objects is proposed to mathematically formulate the obstacles and UGVs using multiple uniform circles. These representations are then converted into convex-concave constraints. After that, the collision-aware obstacle-avoidance problem is formulated as a classical energy minimization problem that can be efficiently solved via a proposed DCCP-based algorithm.

- Both computer simulation and experiment results confirm the effectiveness of the proposed path planning algorithm as compared to the conventional heuristic algorithms such as the Rapidly Exploring Random Tree (RRT) algorithm and Dijkstra methods.

The remainder of this paper is organized as follows. In Section II, the related literature on semantic segmentation and path planning is reviewed. Then, the UAV-assisted semantic segmentation is presented in Section III. After that, the energy-minimization problem is formulated as a non-convex optimization problem before a DCCP-based solution is proposed in Section IV. Finally, extensive computer simulation and experiment results are presented in Section VI before conclusion is given in Section VII.

II. RELATED WORKS

A. UAV-Assisted Semantic Segmentation

The goal of semantic segmentation is to segment the image taken by the UAV before recognizing its content. Driven by the rapid advances of the deep learning technology, especially the convolutional neural network (CNN), the semantic segmentation technology based on deep learning has made great progress. Many pioneering works have been proposed in the literature to solve the problem of semantic segmentation [9]. However, such deep learning-based methods require prohibitively expensive computational complexity. As a result, it is impossible to practically implement these algorithms for real-time semantic segmentation.

In the meantime, it has been recently demonstrated that UAVs possess many unique merits such as flexible mobility, low cost, and high portability, which makes UAVs a promising solution to large-area ground survivance [10]. For instance, commercial UAVs have been used to investigate the rice lodging in paddies over a large area by exploiting FCN-AlexNet for semantic segmentation [11]. Compared to the remote sensing images from satellites [12], more detailed information of a higher spatial resolution can be collected by UAVs.

Recently, [13] presents a segmentation model called YOLACT³ that excels in both accuracy and frames per second (FPS) using a single GPU. Furthermore, an improved version of YOLACT called YOLACT++ is developed in [14] to devise a more accurate segmentation model by adding a new mask re-scoring branch and deformable solutions to the backbone network. As YOLACT++ can achieve the highest accuracy at 33.5 FPS in real-time, it has been envisioned as a promising solution to object segmentation operated on UAVs.

B. Collision-Aware Obstacle-Avoidance Path Planning

Generally speaking, the path-planning algorithms for UGVs can be categorized into two groups, namely the graph-based and the ML-based approaches. For the graph-based approach, the geometry-based and the probability-oriented methods have been extensively investigated in the literature. For instance, the most

representative geometry-based methods include the Dijkstra method, the RoadMap-based visibility graph [15], the Voronoi diagram [16], and the exact cell decomposition (ECD) [17]. Furthermore, the most celebrated probability-oriented methods are the probabilistic RoadMap (PRM) [18] and RRT [19]. On the other hand, the ML-based methods, including the genetic algorithm (GA) [20], particle swarm optimization (PSO) [21] and deep learning (DL)-based algorithms, attract more and more attention in recent years. In general, a training procedure is demanded with the assistance of the existing dataset for these ML-based methods.

Extending to the multi-UGV path planning system, collision avoidance among multiple UGVs requires closer examination related to a more complex constraint description. The authors in [22] adopt the wait-strategy, where the UGV of lower priority is ordered to wait temporarily if two UGVs are collision-prone. Similarly, the path of the lower-priority UGV is planned to bypass the other UGVs, avoiding potential collisions. Despite that a predictive perception model is proposed in [23], the system can only predict possible future lane changes of the competing vehicles. In [24], a mixed-integer programming approach is developed by specifically taking into account the real-time collision problem while modeling the trajectory-generation process as constraints. However, due to the restriction of linear programming, these constraints are usually limited by a convex feasible set.

III. UAV-ASSISTED INDOOR AND OUTDOOR SEMANTIC SEGMENTATION

Empowered with the wide-view superiority, UAVs can be effectively employed to control the UGVs on the ground. In the proposed UAV-UGV cooperative system, YOLACT++ is adopted on a UAV to detect and segment ground objects for both indoor and outdoor scenarios.

A. Semantic Segmentation by YOLACT++

1) *Dataset for UAV Imagery*: At present, several public semantic segmentation data sets are available online. However, most of them only focus on the common objects captured from the ground, such as Ms-COCO [25] and Pascal VOC [26]. There are very few high-resolution semantic segmentation datasets based on UAV imagery. To cope with this problem, a dataset has been developed from a vertical view of UAV for semantic segmentation. In the dataset, 100 high-resolution images are labeled with five object classes, namely Turtlebots, Cars, Pedestrians, Suitcases (acting as obstacles) and Roads. The scenes mainly focus on outdoor highways and indoor corridors, which is suitable for the automatic piloting of UGVs.

2) *Real-Time Semantic Segmentation*: Despite the great success of utilizing LiDAR and RGB-D cameras in object detection and segmentation, their high cost makes them prohibitively unaffordable to be equipped on each UGV on the ground. To circumvent this obstacle, a UAV-assisted cooperative system is proposed with one regular UAV being used to realize collision-aware obstacle-avoidance path planning for multiple UGVs. Particularly, YOLACT++ is used to identify objects and

³You Only Look At CoefficientTs

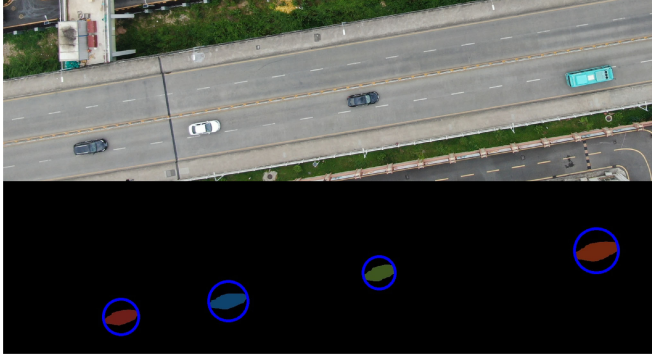


Fig. 3. Illustration of semantic segmentation of an image taken by UAV.

their locations from the pictures taken by the UAV. For various objects, different protection radius are utilized based on the obstacle avoidance strategy of UGV. For instance, if the obstacle is a pedestrian, UGV should be directed farther away from the pedestrian due to the high mobility of human beings.

As mentioned before, YOLACT++ has several practical advantages. Specifically, it is easy to reach 30 FPS even using ResNet-101 [27] such that real-time semantic segmentation can be achieved for automatic piloting. In addition, since the repooling process in YOLACT++ does not incur any image quality loss, the masks generated by YOLACT++ for large objects have a significantly better quality as compared to the more conventional methods. Motivated by these aforementioned advantages, we will use YOLACT++ on the UAV to realize real-time semantic segmentation.

B. Semantic Segmentation and Object Extraction for Outdoor and Indoor Scenarios

In the proposed UAV-UGV cooperative system, the UAV first takes pictures of the ground environment by the wide-angle camera mounted on the UAV. Subsequently, the ground information including UGVs and all possible obstacles along the available road is obtained using semantic segmentation. Then the region of interest (ROI) is mapped to the original image, where the pixels outside the ROI are set to zeros. For the i -th pixel in an image, the following ROI indication function is used to define the ROI area:

$$I(i) = \begin{cases} 1, & \text{if } i \in \text{ROI}, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

where $I(i)$ represents a binary value for the i -th pixel.

Despite the fact that the relative coordinates of an object can be derived from an image, its actual physical location is required for the subsequent path-planning process. Thus, detailed discussions on the location estimation of an object for indoor and outdoor scenarios will be provided in the following section.

1) *Outdoor Scenarios*: As shown in Fig. 3, the UGVs (cars) and obstacles are detected by YOLACT++ and the ROI area is subsequently extracted. For the outdoor environment, GPS is available to provide the accurate position of the UAV. Subsequently, based on the location of the UAV, obstacles and UGVs can be localized by the camera equipped on the UAV as follows. Mathematically, the horizontal position of a UAV can be denoted

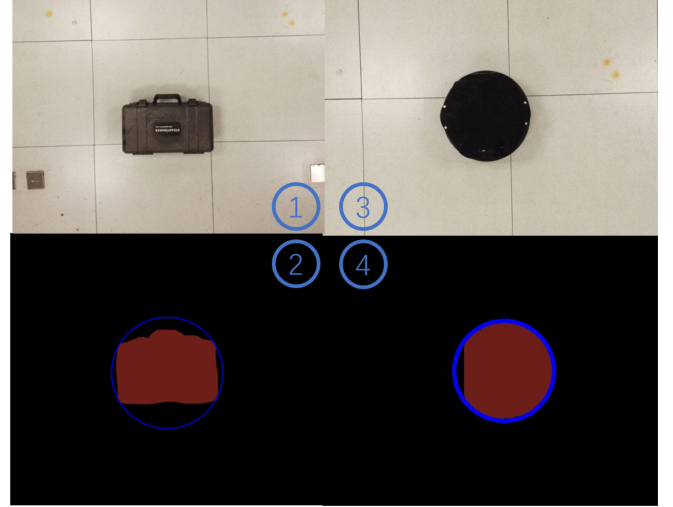


Fig. 4. Object semantic segmentation results. ① and ③ are a suitcase (obstacle) and a Turtlebot (UGV) while ② and ④ are the corresponding segmented results, respectively.

as $\mathbf{p}_{\text{UAV}} = [x_u, y_u]$ while its height as h_u . Thus, the 2-D position of an object can be represented as follows [28]

$$\mathbf{p} = \mathbf{p}_{\text{UAV}} + f_c \cdot \frac{\mathbf{d}_{\text{img}}}{h_u}, \quad (2)$$

where $\mathbf{p}$ is the estimated 2-D position of an object, f_c is the focal of the UAV's camera, and $\mathbf{d}_{\text{img}}$ is the relative pixel distance from the center of an image. The derivation of f_c is given in Section III-C.

Note that the image distortion problem is neglected in this work since the object size is relatively large. The investigation on the distortion error can be found in [29].

2) *Indoor Scenarios*: For the GPS-denied indoor scenarios such as a corridor, the height of the UAV is required to be pre-measured for the coordinate calculation of ground objects. To address this challenge, we propose to add a sensor named TFmini Plus on the UAV to measure its flight height [30]. Despite that its absolute position is not available without the support of GPS, the UAV can measure its relative displacement with high accuracy by Visual Inertial Odometry (VIO) [31]. In this way, the 3-D position of the UAV can be obtained. Again, the actual object size observed in an image can be estimated from Equation (2).

As shown in Fig. 4, an indoor obstacle is detected and extracted as a binary object before the relative coordinate of each pixel is obtained for a given indoor case. After the object detection, the next section will propose an algorithm to estimate the actual object size for given parameters of the UAV's camera.

C. Size Estimation of Ground Objects

Next, the image sequence taken by the UAV is used to estimate the actual size of detected objects in the sequel. This task is achieved by calculating the 3-D coordinates of the corner points of a detected object through the principle of pinhole imaging model [28]. As shown in Fig. 5, the camera of the UAV is located at the optical center, denoted by o , while three coordinate

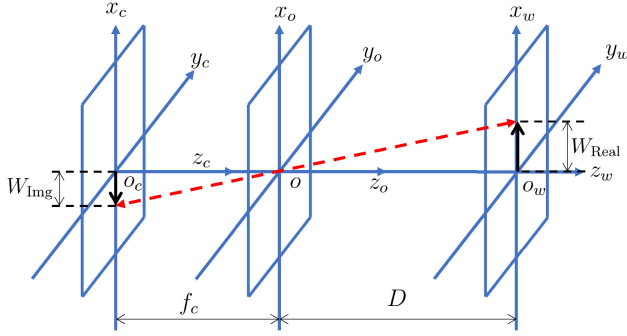


Fig. 5. The pinhole imaging model for size estimation of an object.

systems are established for the camera, the object in the real world, and the image observed, respectively.

1) *Focal Length of the Camera Equipped on the UAV*: We first present the calculation for the focal length of a camera. As illustrated in Fig. 5, the light emitted by an object passes through the optical center of camera o , and subsequently projects onto the image plane. In this process, the object and its image form a triangle similar to the path of light. W_{Img} is the pixel size in the image while D is the distance between the object and the camera.

As W_{Img} and D are two parameters that can be directly measured, only f_c is required for the calculation of W_{Real} . Given a reference object of a given size, the following formulation can be used to derive the focal length from an image taken by a UAV:

$$f_c = \frac{1}{Q} \sum_{q=1}^Q \frac{W_{\text{Img},q} \times D_q}{W_{\text{Real},q}}, \quad (3)$$

where Q measurements are used to reduce the random noise and the subscript q represents the q -th measurement for parameter W_{Img} , W_{Real} and D . With the given f_c , the real size measurement of an object will be given in the following part.

2) *Vision-Based Size Estimation*: For a given image, the distance D between an object and the optical center o is assumed to be known. The focal distance f_c is derived from Equation (3) while the pixel width W_{Img} of an object is measured from the image. The real size of an object can be estimated by:

$$W_{\text{Real}} = \frac{W_{\text{Img}} \times D}{f_c}. \quad (4)$$

The estimation accuracy is verified through extensive experiments shown in Section VI-B.

IV. MATHEMATICAL FORMULATION FOR UAV-UGV COOPERATIVE SYSTEMS

To realize the energy-efficient path planning, a non-convex problem is formulated in which the detected objects are modeled as convex-concave constraints in a collision-aware optimization problem.

A. Shape Representation of Ground Objects

The ground objects detected and segmented above may exhibit any arbitrary 2-D shapes perceived by the UAV. To facilitate

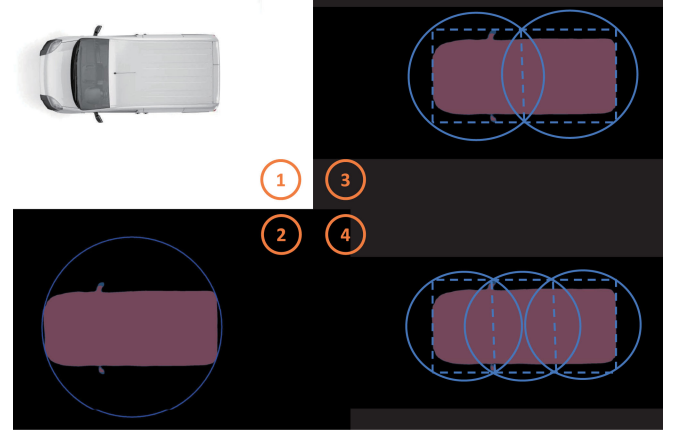


Fig. 6. Reshaping procedure for a car with $K_{\text{max}} = 3$. ① is the original figure. ②, ③ and ④ are the circular representations with $K = 1, 2, 3$ circles corresponding to IoU = 64.7%, 77.1%, 79.7%, respectively.

the formulation of the path-planning optimization problem, the 2-D shape of the ground objects has to be first represented in terms of regular geometric functions. Conventionally, polygon functions can be used to approximate any shapes [32]. For instance, if an obstacle is approximated by a rectangle of width W and length L with $L \geq W$, then the corresponding constraints for path planning are represented as

$$|x_t - x_p| \geq \frac{L}{2} \quad \text{OR} \quad |y_t - y_p| \geq \frac{W}{2}, \quad (5)$$

where (x_t, y_t) are the coordinates of the planned trajectory at the t -th instant while the rectangle obstacle is centered at (x_p, y_p) . In the approximation above, it is assumed that the long side and short side of the rectangle are along with x axis and y axis, respectively. As shown in [32], the constraints in Equation (5) are a set of inequalities with the logical OR operation. Unfortunately, these constraints make the resulting path-planning optimization problem analytically intractable.

In this work, the circular functions is proposed to represent the shapes of the ground objects. For an arbitrary shape, K circular functions can be used to approximate the 2-D object shape as shown in Fig. 6. The most important advantage of using the circular functions is to provide a smooth protective region for collision avoidance while significantly simplifying the following optimization problem formulation, which will become more evidential in the later discussions. To quantify the approximation error between the actual object and its circular representation, the following Intersection over Union (IoU) is proposed [33]:

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}, \quad (6)$$

where “Area of Overlap” refers to the area that the K circles overlap with the object while “Area of Union” is the area of the union of K circles. Clearly, a larger IoU stands for a more accurate approximation.

Given the 2-D shape of a detected object, an algorithm is devised to determine the minimum number of circles that satisfy $\text{IoU} \geq \lambda$ where λ is the minimum required IoU. Since each circle is to be converted into one constraint in the subsequent

Algorithm 1 : Circular Approximation of A Ground Object**Input:**

K : The number of circles;
 I_o : The number of pixels of an extracted object;
 I_u : The number of pixels in the Area of Union with
 $K = 1$;
 λ : The minimum required IoU;
 $K_{\max}$: The maximum number of K ;

Procedures

- 1: $K=1$;
- 2: Calculate $\text{IoU}^{(1)} = I_o/I_u$;
- 3: Find the minimum enclosing rectangle of size $L \times W$ for the object;
- 4: **while** $\text{IoU}^{(K)} < \lambda$ or $K < K_{\max}$ **do**
- 5: Update $K \leftarrow K + 1$;
- 6: Divide the rectangle into K parts uniformly;
- 7: Calculate the radius $r^{(K)}$ using Equation (7);
- 8: Update the I_u for K circles;
- 9: Recalculate $\text{IoU}^{(K)}$;
- 10: **end while**
- 11: Outputs: The center coordinates and radius of K^* circles.

path-planning optimization, a larger number of circles required implies more resulting constraints, *i.e.* higher complexity for the subsequent optimization problem. Thus, the design goal is to find the minimum number of circles.

As detailed in **Algorithm 1**, the propose algorithm first finds the smallest rectangle of size $W \times L$ that can encompass the whole object as shown in Fig. 6. After that, K identical circles are designed to replace the rectangle by satisfying the minimum IoU requirement. More specifically, we begin with $K = 1$ circle whose radius is given by

$$r^{(K)} = \frac{1}{2} \sqrt{(L/K)^2 + W^2}. \quad (7)$$

The proposed algorithm needs to iteratively find the suitable K . For each iteration, the computational complexity is $\mathcal{O}(n^2)$.

Using the IoU definition defined above, the value of K can be incrementally increased until the resulting IoU satisfies $\text{IoU} \geq \lambda$ or K exceeds a predefined threshold $K_{\max}$ given by

$$K_{\max} = \lceil L/W \rceil, \quad (8)$$

where $\lceil \cdot \rceil$ stands for the largest integer not greater than the quantity enclosed.

After obtained the optimal number of circles using **Algorithm 1**, the constraints of an obstacle for path planing can be formulated into simplified convex-concave expressions:

$$\|y_t - p_k\| \geq r^* \quad \text{for } k = 1, 2, \dots, K^*, \quad (9)$$

where y_t is the coordinates of the planned trajectory at the t -th instant with the k -th circle being centered at p_k .

B. Energy-Efficient Path Planning for Multiple UGVs

The collision problem must be considered when multiple UGVs are deployed in the system. The movement of N UGVs can be modeled as N linear dynamic systems. The state model can be formulated as [34]

$$s_{t+1}^{(n)} = A^{(n)} s_t^{(n)} + B^{(n)} u_t^{(n)}, \quad (10)$$

$$y_t^{(n)} = C^{(n)} s_t^{(n)}, \quad (11)$$

where $t = 0, 1, \dots, T-1$ denotes the discrete time while $s_t^{(n)} \in \mathbb{R}^4$ composed of the position and velocity represents the states of the n -th UGV for $n = 1, 2, \dots, N$. Furthermore, $y_t^{(n)} \in \mathbb{R}^2$ stands for the 2-D position output and $u_t^{(n)}$ is the energy consumption required to change the velocity of the n -th UGV at the t -th time instant. For the n -th UGV, the parameter $A^{(n)}$, $B^{(n)}$ and $C^{(n)}$ are defined as following:

$$A^{(n)} = \begin{bmatrix} I_2 & \Delta T I_2 \\ \mathbf{0}_2 & R^{(n)} \end{bmatrix}, B^{(n)} = \begin{bmatrix} \mathbf{0}_2 \\ \Delta T I_2 \end{bmatrix}, C^{(n)} = \begin{bmatrix} I_2 \\ \mathbf{0}_2 \end{bmatrix}^T, \quad (12)$$

where $I_2 \in \mathbb{R}^{2 \times 2}$ is an identity matrix and $\mathbf{0}_2 \in \mathbb{R}^{2 \times 2}$ is a zero matrix. ΔT is the time interval between two consecutive states. In addition, $R^{(n)} \in \mathbb{R}^{2 \times 2}$ is the resistance damping on velocity encountered by the n -th UGV. The value of $R^{(n)}$ depends on the environment such as the friction between the n -th UGV and the ground. For simplicity, $R^{(n)}$ is set as a constant matrix by ignoring the environmental difference in the sequel.

Next, we proceed to solve the collision-aware energy-minimization problem by deriving the optimal velocity profile with emphasis on inter-UGV collision avoidance. Specifically, the constraints for collision avoidance of multiple UGVs are formulated as

$$\|y_t^{(n)} - y_t^{(m)}\| \geq d_{\text{UGV}}, \quad (13)$$

where d_{UGV} is the minimum safety distance between any two UGVs. The constraints for this problem are non-convex, which means that existing methods are not suitable for this problem.

To minimize the total energy consumption, the objective function can be formulated as a fuel minimization problem, *i.e.*,

$$\text{minimize} \quad \sum_{n=1}^N \sum_{t=0}^{T-1} \|u_t^{(n)}\|_1, \quad (14)$$

where $\|\cdot\|_1$ is the ℓ_1 norm of the enclosed vector. Recalling Equation (10), Equation (14) implies that a constant UGV velocity is preferred since velocity variation incurs inevitable fuel consumption based on the Newton's Second Law.

V. ENERGY-EFFICIENT PATH PLANNING USING DCCP

It should be pointed out that the optimization problem in Section IV is non-convex due to the convex-concave constraints. To cope with this problem, a DCCP-based approach is proposed. DCCP is an extension of the difference of convex (DC) programming. Table I summarizes the important symbols used in this section.

TABLE I
SUMMARY OF KEY NOTATIONS

Symbol	Description
$\mathbf{p}_m$	The center position of the m -th circle.
r_m	The radius of the m -th obstacle.
$\mathbf{y}_t^{(n)}$	The center position of the n -th UGV at the t -th instant.
$\mathbf{l}_m$	The closest boundary point for the m -th obstacle.
$\mathbf{u}_t^{(n)}$	The energy consumption of the n -th UGV at the t -th instant.
$P_{\max}$	The maximal power consumption for one instant.
$\mathbf{s}_t^{(n)}$	The position and velocity of the n -th UGV at the t -th instant.
d_m	The minimum distance between a UGV and the m -th obstacle.
d_{UGV}	The minimum safety distance between two UGVs.
Γ	The available road known in advance.

A. A Brief Review of DCCP

Before the DCCP approach is proposed, a short review on DCCP is provided in this section. In general, a DC programming problem can be explicitly designed as:

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} \quad f_0(\mathbf{x}) - g_0(\mathbf{x}) \\ & \text{subject to:} \quad f_i(\mathbf{x}) - g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \end{aligned} \quad (15)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the optimization variable and the functions $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$ and $g_i : \mathbb{R}^n \rightarrow \mathbb{R}$ for $i = 0, \dots, m$ are convex.

The convex-concave procedure (CCCP) is a popular heuristic algorithm for finding a local optimum of Equation (15) [35]. In [7], DCCP is proposed as an extension of CCCP algorithm for DCCP problems formulated based on the DC programming problem defined in Equation (15). More specifically, the DCCP optimization problem takes the following form:

$$\begin{aligned} & \underset{\mathbf{x}}{\text{minimize}} \quad f_0(\mathbf{x}) - \Phi \\ & \text{subject to:} \quad \Phi = g_0(\mathbf{x}), \\ & \quad f_i(\mathbf{x}) \leq g_i(\mathbf{x}), \quad i = 1, \dots, m, \end{aligned} \quad (16)$$

where $\mathbf{x}$ is the original optimization variable in Equation (15) while Φ is a new variable. In this way, the objective function is convex while the constraints can be non-convex. In a DCCP problem, a convex function is optimized subject to non-affine equality and non-convex inequality constraints.

B. Obstacle-Avoidance Path Planning for a Single UGV

We first investigate the obstacle-avoidance path planning for a single UGV, which is modeled as one circle. For simplicity, the superscript (n) is omitted temporarily in the single-UGV case. It is assumed that the generation of a path from the starting position with known velocity $\mathbf{s}_0 = \mathbf{s}_{\text{init}}$ to the destination with required velocity $\mathbf{s}_{T-1} = \mathbf{s}_{\text{end}}$ in the binary image attained in Section III. The obstacles are represented by M circles in the binary image.

Furthermore, it is assumed that the height of ground objects is approximately the same. The center of the m -th circle is denoted by $\mathbf{p}_m$ whose radius is r_m , for $m = 1, 2, \dots, M$. Without loss of generality, it is assumed that the m -th obstacle is closer to the UGV than the n -th obstacle if $m < n$.

Apart from obstacles, the available road denoted by Γ is assumed to be known in advance. Denote by d_m the minimum safety distance required by the m -th circle. To find an available

trajectory, the circles and the road must satisfy at least one of the following requirements for $m = 1, 2, \dots, M$:

$$\begin{cases} C_1 : d_m \leq \|\mathbf{l}_m - \mathbf{p}_m\|_2 - r_m, \\ C_2 : d_m \leq \|\mathbf{p}_{m+1} - \mathbf{p}_m\|_2 - r_{m+1} - r_m, \end{cases} \quad (17)$$

where $\mathbf{p}_m$ stands for the coordinates of the m -th circle for $m = 1, \dots, M$. Furthermore, $\mathbf{l}_m \in \Gamma$ is the closest boundary point for the m -th circle. When a UGV passes by a circle or through two adjacent obstacles, C_1 and C_2 ensure a safe distance between the UGVs and the circle or road boundary nearby.

After the obstacles are detected, the UAV guides its UGV to travel from the starting point to the destination. To ensure the existence of a feasible solution for the following optimization problem, a feasibility check is required to confirm that at least one of the requirements in Equation (17) can be satisfied with the pre-defined parameters.

Denote by $\mathbf{s}_0, \mathbf{s}_1, \dots, \mathbf{s}_{T-1}$ the discrete samples along the planned trajectory, where T is the number of samples within a period. Recalling the derivation in Section IV-B, the obstacle-avoidance energy-efficient trajectory generation for a single UGV can be formulated as the following optimization problem:

$$\begin{aligned} & \underset{\mathbf{u}_t}{\text{minimize}} \quad \sum_{t=0}^{T-1} \|\mathbf{u}_t\|_1 \\ & \text{subject to:} \quad C_1 : \mathbf{s}_0 = \mathbf{s}_{\text{init}}, \mathbf{s}_{T-1} = \mathbf{s}_{\text{end}}; \\ & \quad C_2 : \mathbf{s}_{t+1} = \mathbf{A}\mathbf{s}_t + \mathbf{B}\mathbf{u}_t; \\ & \quad C_3 : \mathbf{y}_t = \mathbf{C}\mathbf{s}_t; \\ & \quad C_4 : \|\mathbf{u}_t\|_{\infty} \leq P_{\max}; \\ & \quad C_5 : \|\mathbf{y}_t - \mathbf{y}_{t-1}\|_2 \leq \gamma; \\ & \quad C_6 : \|\mathbf{y}_t - \mathbf{p}_m\|_2 - r_m \geq d_m; \\ & \quad C_7 : \mathbf{y}_t \in \Gamma, \end{aligned} \quad (18)$$

where $\mathbf{u}_t$ is the instantaneous energy consumption while $P_{\max}$ stands for the maximal power for one instant. $\|\cdot\|_{\infty}$ is the ℓ_{∞} norm of the enclosed quantity. Furthermore, γ is a pre-defined parameter to control the smoothness of the resulting path since γ constrains the maximal distance between two generated trajectory points. More specifically, a smaller γ can lead to a smoother trajectory at the cost of increased computational complexity. Thus, the value of γ must be carefully chosen according to the real environment. In constraint C_6 , the obstacles are detected and segmented by the UAV. Based on **Algorithm 1**, the obstacles are approximated and represented as circles that can be converted into convex-concave constraints.

It is straightforward to show that the problem in Equation (18) is non-convex due to its non-convex constraints. However, since Equation (18) follows the DCCP formulation shown in Equation (16), Equation (18) can be solved under the DCCP framework [7]. In contrast to the conventional trajectory-generation algorithms such as Dijkstra and RRT, the proposed DCCP-based method can formulate an optimization problem with given non-convex constraints. It should be pointed out that the solution provided by the DCCP is shown to be a local optimum [7].

C. Collision-Aware Obstacle-Avoidance Path Planning for Multiple UGVs

In the sequel, we consider the optimization problem for a system consisted of $N > 1$ UGVs. At time instant t , each UGV is required to keep a safety distance of at least d_{UGV} from each other. In addition, the design goal is to minimize the total power consumption. More specifically, given the initial state $\mathbf{s}_0^{(n)} = \mathbf{s}_{\text{init}}^{(n)}$ and the terminal state $\mathbf{s}_{T-1}^{(n)} = \mathbf{s}_{\text{end}}^{(n)}$ for each UGV, $\sum_{n=1}^N \sum_{t=0}^{T-1} \|\mathbf{u}_t^{(n)}\|_1$ is used to calculate the total power consumption of all UGVs over all time instants with limited energy input $\|\mathbf{u}_t^{(n)}\|_\infty \leq P_{\text{max}}$ at one time instant. Thus, the optimization problem can be formulated as

$$\begin{aligned}
 & \text{minimize} \quad \sum_{n=1}^N \sum_{t=0}^{T-1} \|\mathbf{u}_t^{(n)}\|_1 \\
 & \text{subject to: } C_1 : \mathbf{s}_0^{(n)} = \mathbf{s}_{\text{init}}^{(n)}, \mathbf{s}_{T-1}^{(n)} = \mathbf{s}_{\text{end}}^{(n)}; \\
 & \quad C_2 : \mathbf{s}_{t+1}^{(n)} = \mathbf{A}^{(n)} \mathbf{s}_t^{(n)} + \mathbf{B}^{(n)} \mathbf{u}_t^{(n)}; \\
 & \quad C_3 : \mathbf{y}_t^{(n)} = \mathbf{C}^{(n)} \mathbf{s}_t^{(n)}; \\
 & \quad C_4 : \|\mathbf{y}_t^{(n)} - \mathbf{y}_t^{(j)}\| \geq d_{\text{UGV}}, \quad \text{for } j \neq n; \\
 & \quad C_5 : \|\mathbf{u}_t^{(n)}\|_\infty \leq P_{\text{max}}; \\
 & \quad C_6 : \|\mathbf{y}_t^{(n)} - \mathbf{y}_{t-1}^{(n)}\|_2 \leq \gamma; \\
 & \quad C_7 : \|\mathbf{y}_t^{(n)} - \mathbf{p}_m\|_2 - r_m \geq d_m; \\
 & \quad C_8 : \mathbf{y}_t^{(n)} \in \Gamma. \tag{19}
 \end{aligned}$$

The optimization problem of energy minimization is developed in Equation (19), where constraints C_4 and C_7 are non-convex. Similar to Equation (18), Equation (19) follows the DCCP formulation and can be solved as a DCCP problem by using any existing solvers.

D. Smoothness of Path Planning

A smooth and continuous UGV's trajectory without sharp turns is desired in practice. We will first characterize the trajectory's smoothness. For any three ordered positions $\{\mathbf{y}_{i-1}, \mathbf{y}_i, \mathbf{y}_{i+1}\}$ for $i \geq 1$, the turning angle is defined as :

$$\cos \beta_i = \frac{(\mathbf{y}_{i+1} - \mathbf{y}_i)^T \cdot (\mathbf{y}_i - \mathbf{y}_{i-1})}{\|\mathbf{y}_{i+1} - \mathbf{y}_i\| \cdot \|\mathbf{y}_i - \mathbf{y}_{i-1}\|}, \tag{20}$$

where $\mathbf{v}^T$ stands for the transpose of vector $\mathbf{v}$.

To compare the smoothness of the trajectories passing through I positions generated by different algorithms, the following smoothness metric is employed:

$$S = \sum_{i=1}^{I-2} |\beta_i|. \tag{21}$$

Inspection of Equation (21) suggests that a smaller S implies a smoother path defined by $\{\mathbf{y}_i\}$ for $i = 0, 1, 2, \dots, I-1$. This newly defined smoothness metric will be employed to compare the performance of different algorithms.

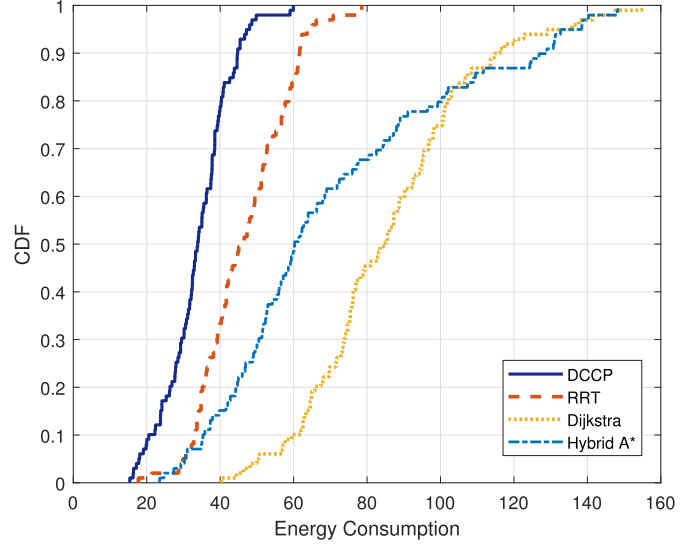


Fig. 7. CDF of energy consumption for different algorithms using 100 random experiments.

VI. COMPUTER SIMULATION AND EXPERIMENT RESULTS

To evaluate the performance of the proposed UAV-UGV cooperative system, both computer simulation and proof-of-concept experiments are performed.

A. Computer Simulation

We first compare the energy efficiency of the proposed DCCP-based algorithm, RRT Dijkstra, hybrid A*[36] via computer simulation. In the computer simulation, 10 obstacles are randomly placed within an area of size 30×30 while the radii of the obstacles are randomly chosen from $r_m \in \{1, 2, 3, 4\}$ for $m = 1, \dots, 10$. One single UGV is considered with random initial and destination positions defined by $\{x_{\text{init}}, y_{\text{init}}\}$ and $\{x_{\text{end}}, y_{\text{end}}\}$, respectively, with $x_{\text{init}} \in [0, 16]$, $y_{\text{init}} \in [0, 10]$, $x_{\text{end}} \in [16, 30]$ and $y_{\text{end}} \in [22, 30]$. The cumulative distribution function (CDF) of the energy consumption over 100 computer runs is plotted in Fig. 7. Inspection of Fig. 7 indicates that the energy consumption of the DCCP-based algorithm is more efficient as compared to that of RRT and Dijkstra algorithms. This is because of the fact that the objective function in Equation (19) is formulated as energy minimization.

Next, we proceed to simulate the multi-UGV scenarios. Fig. 8 shows a path-planning example for two UGVs. To better illustrate the collision avoidance capability, Fig. 8 shows the two trajectories generated with the proposed DCCP-based scheme over three time slots. In sharp contrast to the conventional wait-strategy scheme reported in [22], the proposed algorithm optimizes the trajectories globally for both UGVs. To reach their destinations, two UGVs are steered to enter the collision area in different time slots, which ensures that the UGVs can avoid colliding with each other. As formulated in Equation (19), C_6 constrains the minimal distance between two UGVs using the velocity control. Meanwhile, the ground obstacles surrounding the UGVs are also identified as shown in Fig. 8. The planned

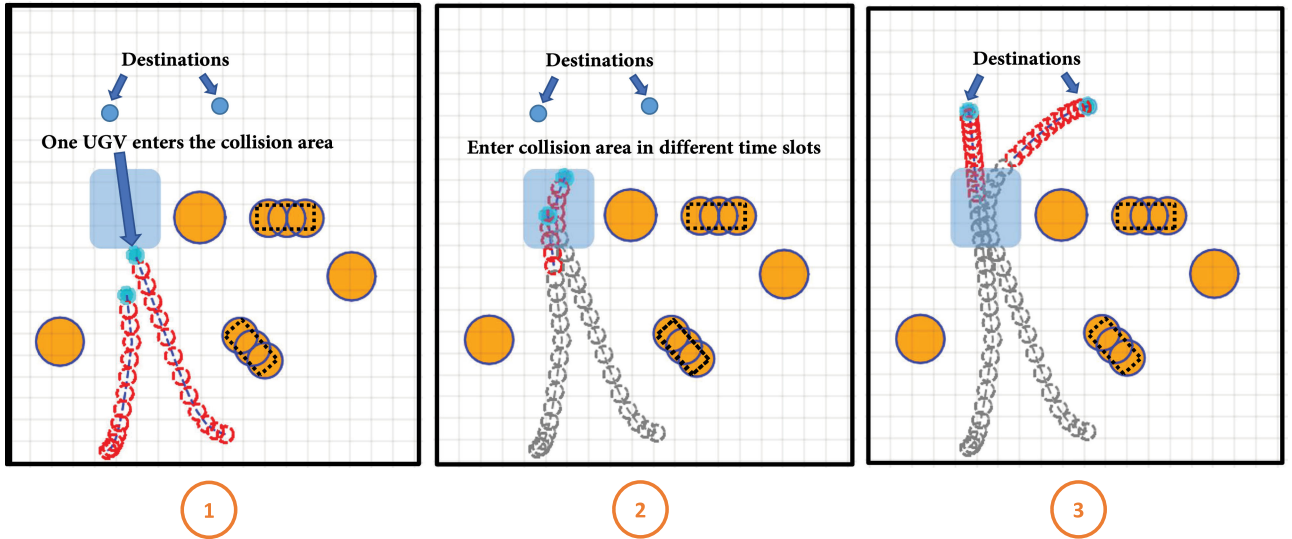


Fig. 8. Collision-aware obstacle-avoidance path planning for two UGVs. ① One UGV is faster than the other one. Thus, it is entering the collision area while the other UGV is still far away from the collision area. ② Two UGVs enter the collision area in different time slots. Thus, the collision problem is solved. ③ Two UGVs pass the collision area and go to their destinations.

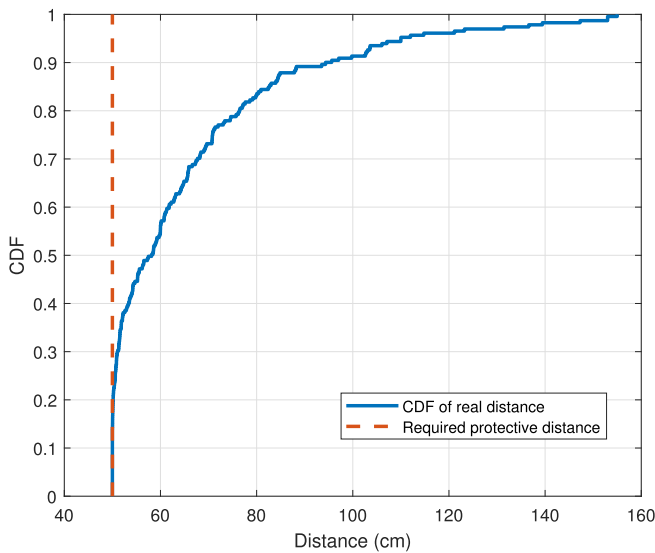


Fig. 9. CDF of distance between two UGVs.

UGV trajectories effectively detour from the obstacles and avoid collisions.

To explicitly evaluate the capability of the proposed system in assisting UGVs to avoid collisions, the CDF of distance between two UGVs are illustrated in Fig. 9, where the required safety distance is set to be $d_{UGV} = 50$ cm for two UGVs in Equation (19). Fig. 9 shows that the resulting distance between UGVs exceeded d_{UGV} during the entire trajectories leading to their destinations.

B. Proof-of-Concept Experiments

In this section, a proof-of-concept testbed is constructed using a Turtlebot2 as a UGV and a Mavic Air from DJI as the UAV. Fig. 10 depicts the setup of our proof-of-concept

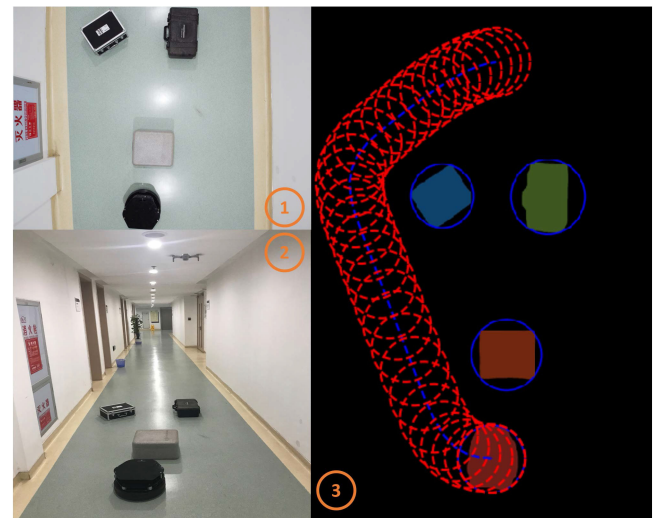


Fig. 10. Experiment setup with three obstacles, one Turtlebot2 and one Mavic Air. ① The picture captured by the UAV's built-in camera. ② An example of UAV-UGV cooperative system for indoor scenarios. ③ Obstacle-avoidance path planning using the semantic segmentation and DCCP-based algorithm.

experiment. The top-left figure in Fig. 10 shows the top view taken by the UAV, showing three obstacles and one Turtlebot2. The obstacles and the Turtlebot2 are then detected and extracted as a binary image as shown in the right figure in Fig. 10. After that, the proposed DCCP-based scheme generated the energy-efficient path bypassing the obstacles.

First, we investigate the accuracy of the vision-based estimation proposed in Section III-C2. A pre-trained model *Darknet53_54_800000* is used to segment the objects using YOLACT++ [13]. The real-time 33.5 FPS semantic segmentation is performed. Fig. 11 shows the estimated size of two objects as a function of the UAV's flight altitude ranging from

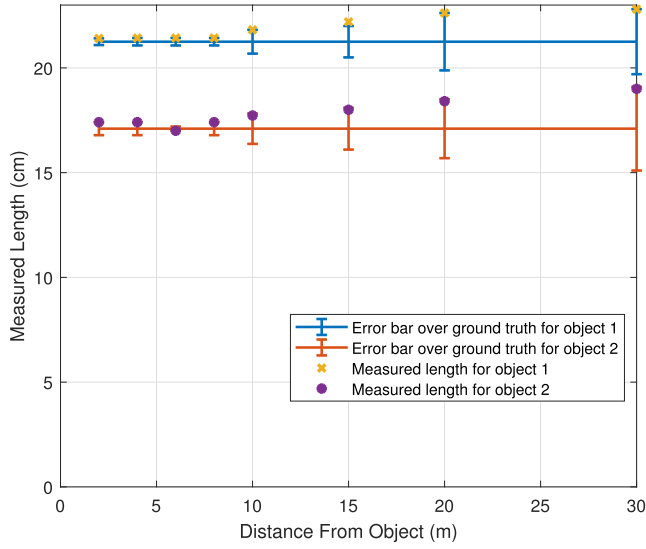


Fig. 11. Size estimation based on UAV's view.

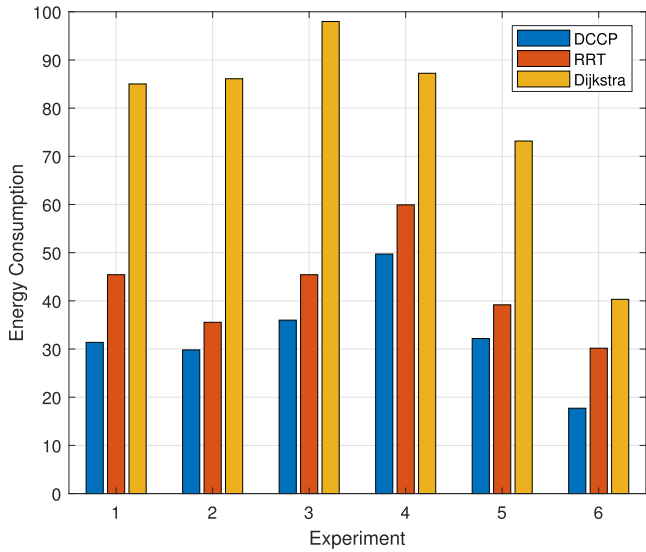


Fig. 12. Energy-consumption comparison for different algorithms.

2 m to 30 m. Inspection of Fig. 11 reveals that the estimation error is around 3% of the actual size, regardless of the UAV's flight altitude.

Next, six experiments are set up to compare the energy consumption defined in Equation (19) for the proposed algorithm and other conventional algorithms, *i.e.*, RRT and the Dijkstra algorithms over six experiments in which obstacles are randomly placed. Two UGVs are used and the number of obstacles ranges from 1 to 3. In these experiments, the initial and terminal points are randomly selected. Since the proposed algorithm is specifically designed to minimize the energy consumed by the UGVs, it is not surprising to observe that the proposed algorithm can achieve the best energy performance in all six experiments as shown in Fig. 12.

The higher energy efficiency of the proposed algorithm can be interpreted from the following comparisons on path length and smoothness. Fig. 13 shows the path lengths of different algorithms in all six experiments. Clearly, the proposed DCCP-based

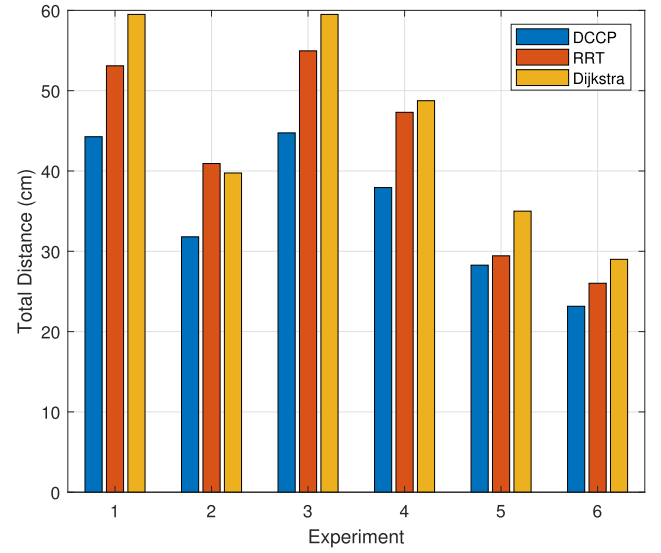


Fig. 13. Path length comparison for different algorithms.

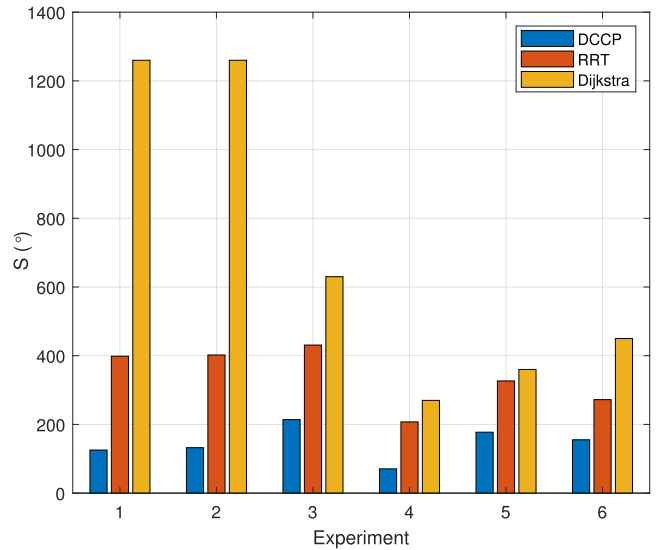


Fig. 14. Smoothness comparison for different algorithms.

algorithm has the shortest path length in all six experiments. This can be explained from the following intuition. Since both Dijkstra and RRT algorithms first divide the space into grids, their trajectories are discretized and make many sharp 90° turns. In sharp contrast, the proposed DCCP-based algorithm derives the trajectory without first discretizing the space as formulated in Equation (19). As a result, the DCCP-based path has a shorter length with less energy consumption.

Finally, Fig. 14 compares the smoothness of the trajectories generated by the proposed algorithm, RRT and Dijkstra algorithms in the six experiments. Inspection of Fig. 14 reveals that the proposed algorithm can generate a much smoother trajectory in terms of the smoothness metric defined in Equation (21). It should be pointed out that the path generated by the proposed algorithm requires less energy consumption due to its shorter length and better smoothness.

VII. CONCLUSION AND FUTURE WORK

In this work, a UAV-UGV cooperative system has been developed to achieve UAV-assisted path planning for UGVs. The UAV in the system first performs object detection via semantic segmentation to extract information of the ground obstacles. After that, the ground obstacles are represented by circular approximation before an energy-minimization path planning optimization problem for UGVs is formulated. Capitalizing on the DCCP technique, an algorithm has been proposed to derive the optimal collision-aware obstacle-avoidance trajectories for UGVs. Proof-of-concept experiments have been conducted to validate the performance of the proposed algorithm. Both computer simulation and experiment results have shown that the UGVs can effectively avoid collisions while achieving high energy efficiency using the path generated by the proposed path-planning algorithm.

In our future work, realistic dynamic obstacles will be considered to develop more practical obstacle-avoidance path-planning algorithms. In addition, the mathematical formulation for the obstacles can be further improved to increase the IoU. The mobility problem of the UAV is neglected in this work, which is indeed importance for the UAV to track the UGVs during the trajectory generation process. Thus, the path planning for the UAV will be also taken into consideration in our future work.

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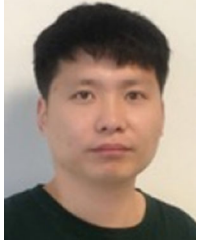
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